Imp Ques

ch 1- introduction.pdf
ch 2- fundamentals of DL.pdf
Ch 3- regularization and optimization.pdf

PART 1 — Exam Ready 10-Mark Answers

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1. Backpropagation Algorithm (10 Marks)

Definition

Backpropagation is a supervised learning algorithm used to train feedforward neural networks by **minimizing the loss function using gradient descent**.

It computes gradients of weights by **propagating errors backward through the network**.

Steps of Backpropagation

1. **Initialize Weights**
    - Assign small random values to weights and biases.
2. **Forward Propagation**
    - Input data is passed through the network.
    - Each neuron computes

$$z = \sum w_i x_i + b$$

- Activation function generates output.

3. **Compute Loss**

- Compare predicted output with actual output.

Example loss:

$$L = \frac{1}{2}(y - \hat{y})^2$$

4. **Backward Propagation**
   - Compute gradient of loss w.r.t weights using chain rule.
5. **Update Weights**

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

where
η = learning rate

6. **Repeat**
   - Continue for all training samples until convergence.

Role in Deep Learning

- Enables efficient training of deep networks
- Reduces prediction error
- Optimizes model parameters

Advantages

- Efficient gradient computation
- Works for multi-layer networks
- Scalable for deep learning

Limitations

- Sensitive to learning rate

- Can suffer from vanishing gradient

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2. Activation Functions (10 Marks)

Definition

An activation function determines the **output of a neuron by introducing non-linearity** into the neural network.

Without activation functions, neural networks behave like **linear models**.

Need for Activation Functions

1. Introduce non-linearity
2. Enable complex pattern learning
3. Control output range
4. Improve decision making ability

Important Activation Functions

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1. Sigmoid Function

$$f(x) = \frac{1}{1 + e^{-x}}$$

Range: **0 to 1**

Advantages

- Interpretable as probability

Disadvantages

- Vanishing gradient problem

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2. Tanh Function

$$tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Range: **-1 to 1**

Advantages

- Zero-centered output

Disadvantages

- Still suffers from vanishing gradient

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3. ReLU

$$f(x) = max(0, x)$$

Advantages

- Fast training
- Avoids vanishing gradient

Disadvantages

- Dying ReLU problem

Comparison

Function	Range	Problem
Sigmoid	0–1	Vanishing gradient
Tanh	-1–1	Saturation
ReLU	0–∞	Dead neurons

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3. Feedforward Neural Network (10 Marks)

Definition

A Feedforward Neural Network (FNN) is a neural network where **information flows only in one direction** from input to output layer.

There are **no cycles or feedback loops**.

Architecture

Layers include:

1. Input Layer
2. Hidden Layer(s)
3. Output Layer

Working

1. Input layer receives data
2. Hidden layer performs computations
3. Weights multiply inputs
4. Activation functions applied
5. Output layer produces prediction

Mathematical Representation

$$y = f(Wx + b)$$

Where

- W = weight matrix
- b = bias
- f = activation function

Advantages

- Simple architecture
- Works for classification and regression

Limitations

- Cannot capture temporal relationships
- Requires large datasets

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4. Vanishing and Exploding Gradient Problem (10 Marks)

Definition

During training, gradients may become extremely **small or extremely large**.

This causes difficulty in updating weights.

Vanishing Gradient

Gradient approaches **zero** during backpropagation.

Effects

- Slow learning
- Earlier layers stop learning

Exploding Gradient

Gradient becomes **very large**.

Effects

- Unstable training
- Large weight updates

Causes

1. Deep networks
2. Sigmoid activation
3. Poor weight initialization

Solutions

1. ReLU activation
2. Batch normalization
3. Gradient clipping
4. Proper initialization

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5. Stochastic Gradient Descent (10 Marks)

Definition

Stochastic Gradient Descent (SGD) is an optimization algorithm that updates model parameters using **one training sample at a time**.

Algorithm Steps

1. Initialize weights randomly.
2. For each training example:

Compute prediction.

3. Calculate loss.
4. Compute gradient.
5. Update weights:

$$w = w - \eta \nabla L(w)$$

6. Repeat for all epochs.

Advantages

- Faster training
- Suitable for large datasets

Disadvantages

- Noisy updates

- Requires learning rate tuning

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6. L1 and L2 Regularization (10 Marks)

Regularization reduces **overfitting** by penalizing large weights.

L1 Regularization

Also called **Lasso regression**.

Loss function:

$$L = Loss + \lambda \sum |w|$$

Effect

- Encourages sparse weights
- Some weights become zero

L2 Regularization

Also called **Ridge regression**.

Loss function:

$$L = Loss + \lambda \sum w^2$$

Effect

- Shrinks weights toward zero
- Prevents large weight values

Comparison

Feature	L1	L2
Penalty	Absolute	Squared
Effect	Sparse weights	Small weights
Use	Feature selection	Generalization

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7. Dropout Regularization (10 Marks)

Definition

Dropout is a regularization technique where **random neurons are temporarily removed during training**.

Working

1. During training
2. Random neurons are disabled
3. Remaining neurons train the network
4. Different neurons drop each iteration

Benefits

- Reduces overfitting
- Improves generalization
- Acts like ensemble learning

Dropout Probability

Typical value

p = 0.5

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8. Data Augmentation (10 Marks)

Definition

Data augmentation increases dataset size by creating **modified versions of existing data**.

Examples

For images

- Rotation
- Flipping
- Cropping
- Scaling
- Noise addition

Advantages

1. Improves model generalization
2. Prevents overfitting
3. Works well with small datasets

PART 2 — Ultra Fast 2–3 Mark Definitions

Memorize these.

Machine Learning

A field of AI that enables systems to learn patterns from data without explicit programming.

Likelihood

Likelihood measures how probable observed data is given a model.

Posterior Probability

$P(H|D)$

Probability of hypothesis after observing data.

Gradient Descent

An optimization technique used to minimize loss function by updating parameters iteratively.

Feedforward Network

A neural network where information flows only from input to output layer.

ReLU

$$f(x) = max(0, x)$$

Activation function widely used in deep networks.

Overfitting

When a model performs well on training data but poorly on unseen data.

Regularization

Technique used to reduce overfitting by penalizing model complexity.

Dropout

Technique that randomly disables neurons during training.

Data Augmentation

Process of increasing dataset size by modifying existing samples.

PART 3 — 6-Page LAST MINUTE REVISION SHEET

Study this before exam.

Page 1 — Neural Network Basics

Neuron equation

$$y = f(\sum w_i x_i + b)$$

Components

- Weights
- Bias
- Activation function
- Loss function

Page 2 — Activation Functions

Function	Formula
Sigmoid	1/(1+e^-x)
Tanh	(e^x-e^-x)/(e^x+e^-x)
ReLU	max(0,x)
Softmax	probability distribution

Page 3 — Neural Network Architecture

Layers

Input → Hidden → Output

Key ideas

- Feature learning
- Nonlinearity

- Parameter optimization

Page 4 — Training Process

1 Forward pass
2 Loss computation
3 Backpropagation
4 Weight update

Page 5 — Optimization

Algorithms

- Gradient Descent
- Stochastic Gradient Descent
- Mini-batch Gradient Descent

Page 6 — Regularization

Techniques

- L1 regularization
- L2 regularization
- Dropout
- Data augmentation
- Early stopping

FINAL SUPER IMPORTANT QUESTIONS

Most likely exam questions:

1. Backpropagation algorithm
2. Activation functions
3. Feedforward neural network
4. Vanishing gradient problem
5. Stochastic gradient descent
6. L1 vs L2 regularization
7. Dropout technique
8. Data augmentation
9. Supervised vs Unsupervised learning
10. Resampling methods

Syllabus comparison

IT Dept (Img 2)

Module No.	Unit No.	Details	Hrs.	CO
1	**From Machine Learning to Deep Learning**		10	CO1
	1.1	Math Behind Machine Learning: Statistics, Probability Conditional Probabilities, Posterior Probability, Distributions, Samples Versus Population, Resampling Methods, Selection Bias, Likelihood		
	1.2	Bayesian Statistics, Supervise and Unsupervised Learning		
	1.3	Stochastic Gradient Descent		
	1.4	Building a Machine Learning Algorithm		
	1.5	Challenges Motivating Deep Learning		
2	**Fundamentals of Deep Networks**		08	CO2
	2.1	Common Architectural Principles of Deep Networks		
	2.2	Deep Feedforward Networks – Example of Ex OR		
	2.3	Gradient-Based Learning		
	2.4	Hidden Units, Architecture Design		
3	**Regularization and Optimization for Training DeepModels**		12	CO2
	3.1	Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness		

AI DS dept portion (Img 1)

1	**Foundations of Neural Networks and Deep Learning**		12	CO1
	1.1	Parametric vs. nonparametric learning, Supervised parametric learning, Unsupervised parametric learning, Nonparametric learning, Transfer Learning		
	1.2	Activation functions, Loss functions, Hyperparameters-Magnitude Hyperparameters, Regularization		
	1.3	Common Architectural Principles of Deep Networks, Building Blocks of Deep Networks, Major architectures of Deep Networks, Advances in network architecture, Generative modelling,		
2	**Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization**		12	CO1
	2.1	Hyperparameter Tuning and Batch normalization- Tuning process, finding optimal Hyperparameters- Learning rate, Batch size, Momentum, Weight decay, Hyperparameter Optimization Algorithms- Grid search, Random search, Bayesian Optimization		
	2.2	Regularization Techniques-L1, L2, Early stopping, Dropout, Data Augmentation		
	2.3	Optimization algorithms- Challenges in Optimization, Mini-batch Gradient Descent, Exponentially Weighted Averages, Gradient Descent with Momentum, AdaGrad, RMSProp, Adam		
3	**Convolutional Neural Networks**		08	CO2
	3.1	•Convolutional Neural Networks, Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior		

Image 2 Topic	Present in Image 1?	Where in Image 1
Statistics & Probability basics	No	Not present
Conditional probability	No	Not present
Posterior probability	No	Not present
Distributions	No	Not present
Samples vs Population	No	Not present
Resampling methods	No	Not present
Selection Bias	No	Not present
Likelihood	No	Not present
Bayesian Statistics	No	Not present
Supervised Learning	Yes	Module 1.1
Unsupervised Learning	Yes	Module 1.1
Stochastic Gradient Descent	Yes	Module 2.3 (Mini-batch Gradient Descent / optimization)
Building ML Algorithm	Partially	Deep network building blocks (1.3)
Challenges motivating Deep Learning	Partially	Advances in network architecture
Common Architectural Principles of Deep Networks	Yes	Module 1.3
Deep Feedforward Networks	Yes	Building blocks of deep networks
Gradient-Based Learning	Yes	Optimization algorithms
Hidden Units / Architecture Design	Partially	Hyperparameters / architecture concepts
Parameter Norm Penalties	Yes	L1, L2 Regularization
Regularization	Yes	Regularization techniques
Dataset Augmentation	Yes	Data augmentation

Noise Robustness	No	Not present
Semi-Supervised Learning	No	Not present
Multi-Task Learning	No	Not present
Early Stopping	Yes	Module 2.2
Parameter Tying / Sharing	No	Not present
Sparse Representations	No	Not present
Bagging / Ensemble Methods	No	Not present
Dropout	Yes	Module 2.2
Adversarial Training	No	Not present
Tangent Distance / Tangent Prop	No	Not present
Manifold Tangent Classifier	No	Not present